



Valve on VR

About 1/3 of gaming mammoth Valve is devoted to VR and the future of their HTC Vive VR system <http://dgit.in/valvevr>



Apple returns money

Consumers are finally getting their money back after Apple was found liable for overpricing books <http://dgit.in/iReturn>

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Q When it comes to the Internet of Things (IoT) the theoretical aspects itself are very exciting to us and our readers. What you're into – implementation at that macro level such as smart cities – is an equally exciting but entirely different ballgame. Can you give us an overview of say what the architecture of such an implementation is likely to be, or what the roadmap is?

Kavitha: When we talk about Smart Cities, it's a bunch of IoT solutions that are coming together, because for something to be smart, it just needs a context. To define context let's think of a bus going in a certain direction. If the bus also has an idea of how many passengers it will likely pick up on the road and how well utilized it will be at the end of the day, it gives a better sense resource management to the administration. Otherwise transport would be running ten loops of the same bus every day without any knowledge of where the density is. Transport is at times underutilised for the simple reason that the last man is not connected. It can happen that the density of the population is elsewhere and the busses are plying in a different direction. So the question to answer is how can I actually get information on that population and the heatmap and marry it with the transportation logistics.

We take solutions from you – the Makers community – put them together and create an architecture around that. Next we also try to do some pattern analytics between multiple data points that are coming in.

At the very basic level there are a bunch of sensors sending in data. Multiple sensors are feeding information and they're correlated. What Intel does is once we have this first level technology built, Intel also has something known as Edge Analytics. You don't have to send all this data back to city centre or a centralised command centre or a data centre because this may be something that could be actually controlled right at the location.

Q What's the advantage in not having it centrally analysed?

Kavitha: Bandwidth and security con-

Building Smart Cities with Intel

We speak to Kavitha Mohammad, Director, IoT & Smart Cities at Intel, Asia Pacific Japan to get a pulse on how Intel plans to build the technological backbone required to support the smart cities of tomorrow.



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cerns. We're creating islands of intelligence. As much as possible if you can control and manage that island, you're actually doing yourselves a favour because you're controlling the data that's moving back and forth, your security concern is to some extent mitigated because it's not going on to a public cloud, and the cost of the bandwidth is also reduced to some extent. So that's the essence of it. A gateway solves the same things that the IoT community has been struggling with. Manageability: how do you manage multiple devices? Interoperability: how do you make sure that the different languages spoken are understood? And then how do you manage Security. These three are very important anywhere in the stack. If you look at the Intel architecture, we have intelligence in the sensors. The Edison that you spoke about for example, it has intelligence built into it. There is also a programmable chip that's coming into play. You don't need to program from a different operating system or console. The chip itself is a FPGA programmable chip. The Gateway handles multiple sensors and it has an operating system built onto it. The operating system will help you program these and it has security and analytics built into it. So the edge analytics island I was talking about earlier is taken care of here. And at the next stage in the stack, many such gateways then come onto a communications network.

Q And these gateways are Intel gateways?

Kavitha: These are intel gateways yes. The OEMs are different, but the underlying chip is Intel.

Q And the OS?

Kavitha: It's a Wind River OS. I'm not sure if you've heard of it but this is the OS that's being used on Mars.

Q How much of our critical infrastructure should we put on the grid?

Kavitha: That's where the security protocols come in. It's a given that there would be one or two corner cases, which might get missed. But the value that you're generating out of putting something on a grid probably far exceeds that. **Q**

(full interview at dgit.in/IntelIoTdg716)